Analytics & Big Data

Session 1: The rise of analytics

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Why is Modern Analytics so Successful?

1. More Data
2. More Computing Power
3. New Algorithms
4. New Analytics Processes

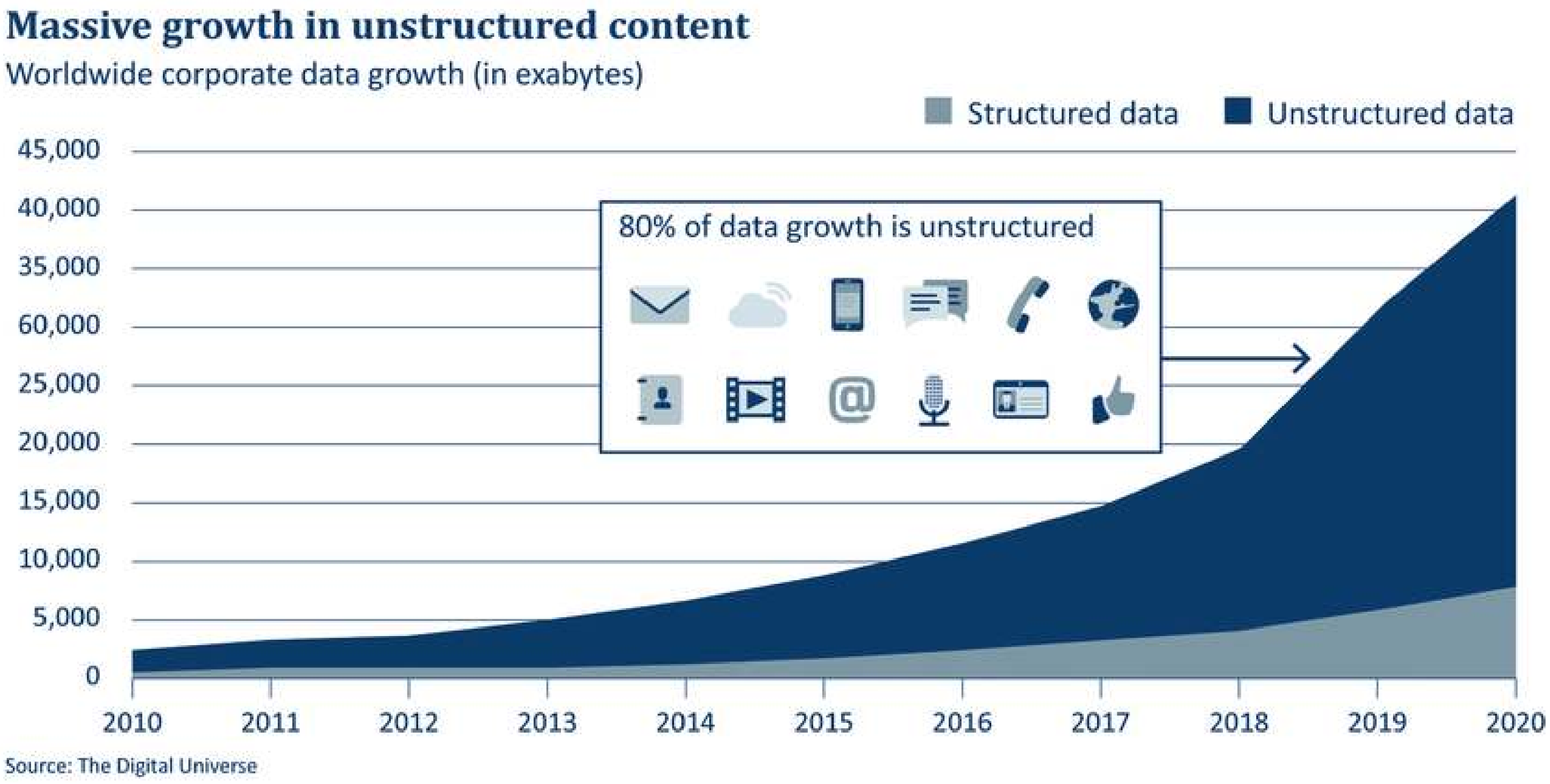
More Data for the Analysis

Internet of Things as a driver of data

Source: Bernard Marr; Source: Ruoming Jin

Social media as a driver of data

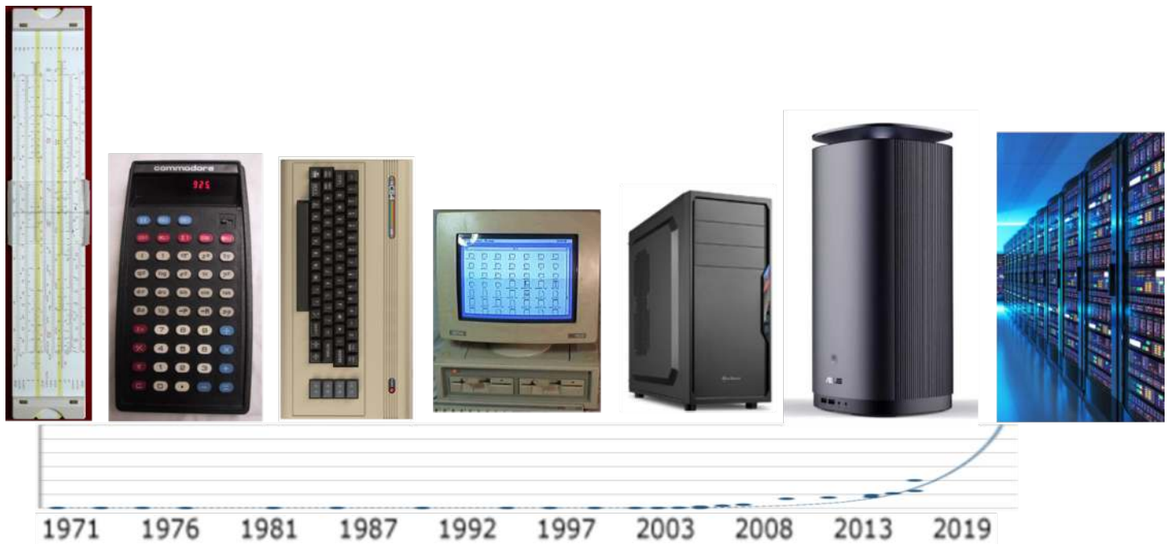
Worldwide Corporate Data Growth



Computing Power

Growth of Computing Power

The rapid acceleration of computing power—driven by advances in hardware, cloud infrastructure, and parallel processing—has enabled modern analytics and machine learning to scale to massive datasets.

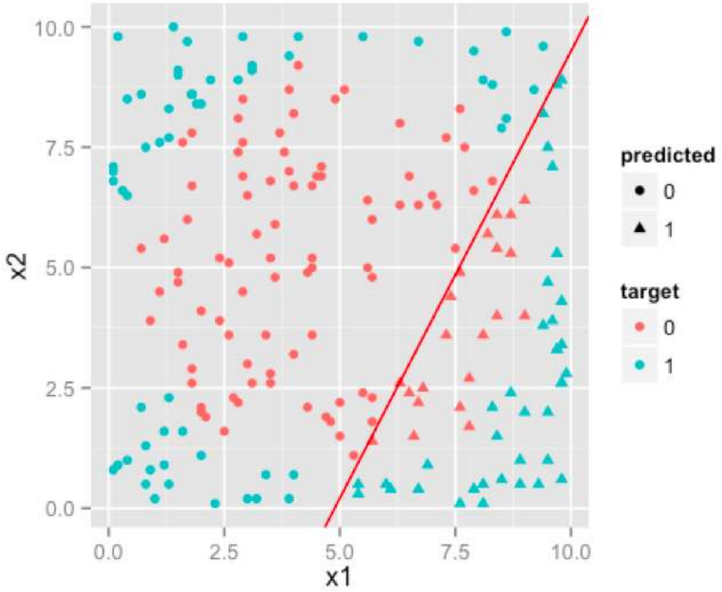


TODO: mention DEA automata. Von Neuman architecture, possible addition of Quantum computing TODO: mention chip miniaturization and pushing pysical limits https://www.youtube.com/watch?v=MiUHjLxm3V0

New Algorithms

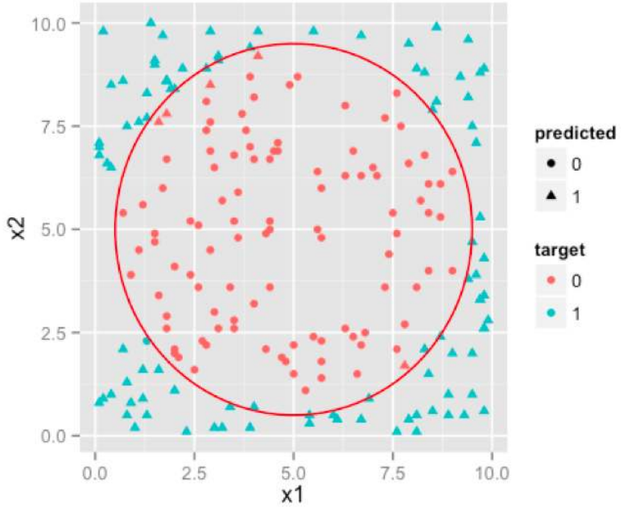
TODO : tBD: should it be “algorithms” or models?

Old vs. New Methods

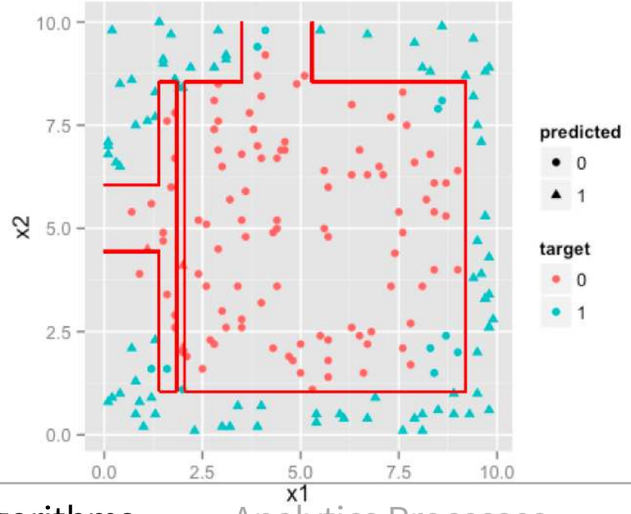


Regression

Traditional



Decision Tree

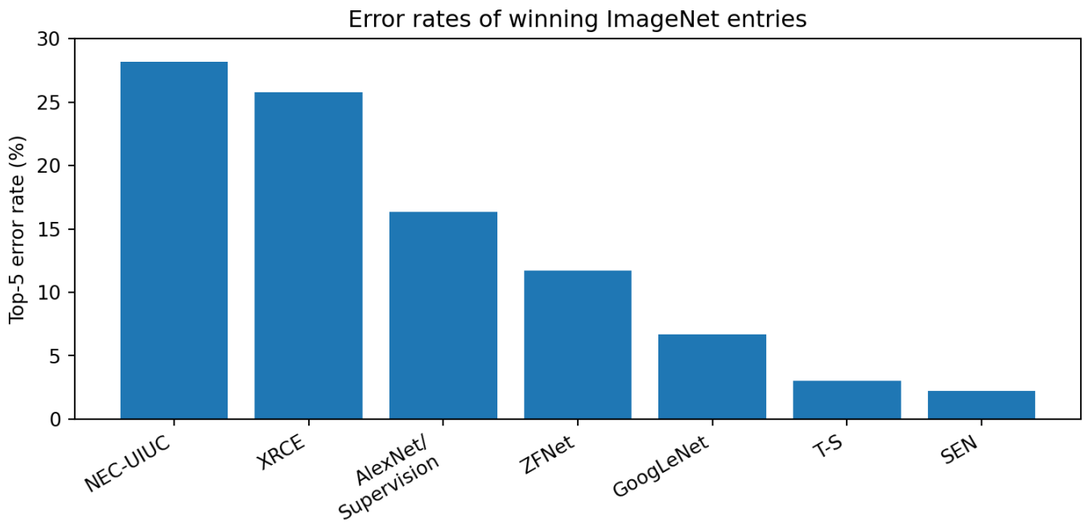


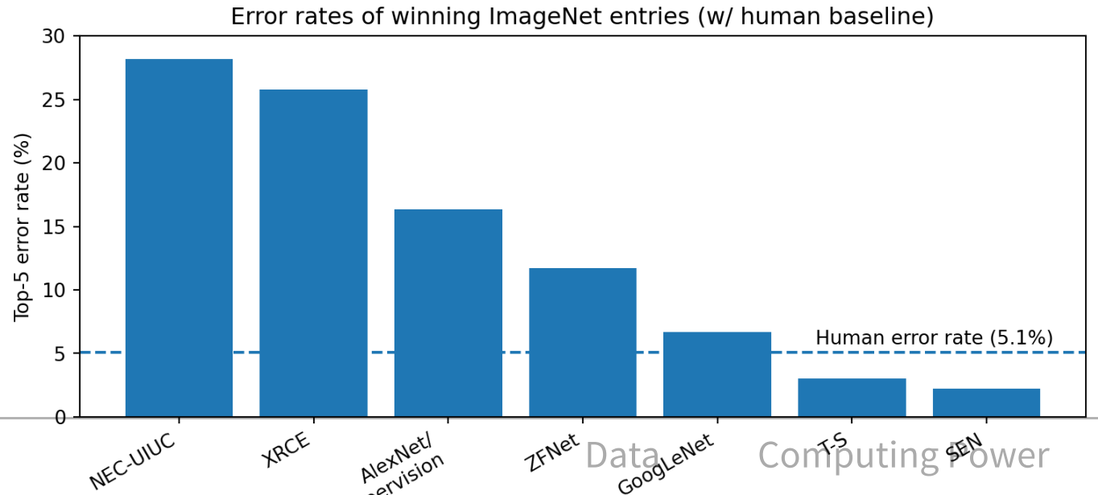
Neural Network

Source: Lalit Sachan

ImageNet

ImageNet is a large-scale image recognition competition where models compete to classify millions of images across thousands of categories. Breakthrough models have historically demonstrated major advances in deep learning.: Models vs Human Baseline

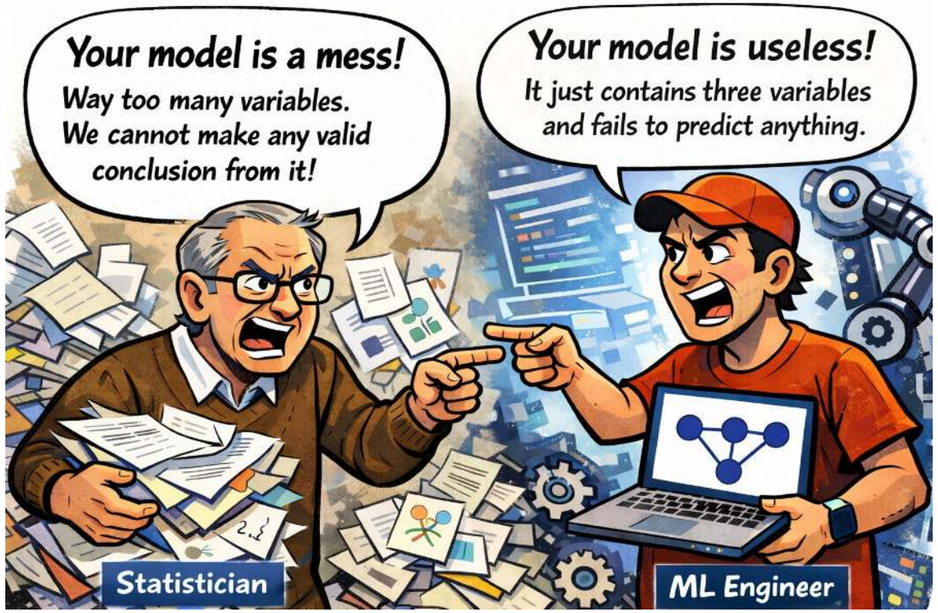




Analytics Processes

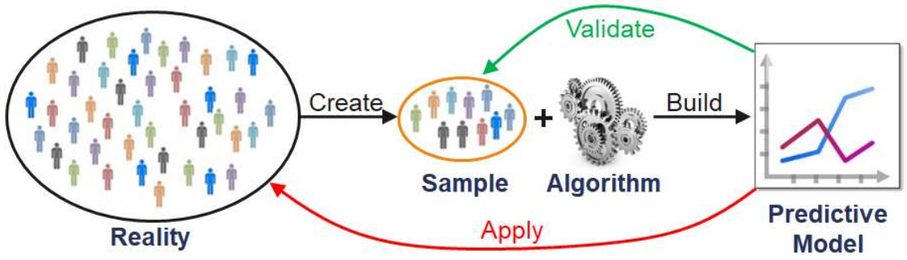
Structure: TODO

- start with epistemic goals (describing, predicting, prescribing)
- introduce CRISP-DM
- distinguish different analytical models and disciplines (referring to the epistemic goals - example: statistician would not be happy with an ML model because it does not support inference/explanation. ML engineer is not happy with a regression model - it may support inference/explanation better, but its predictions may be less accurate)

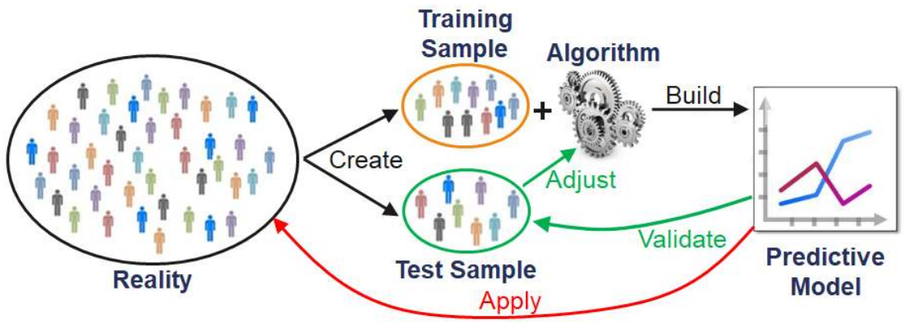


Traditional vs. Modern Analytics Process

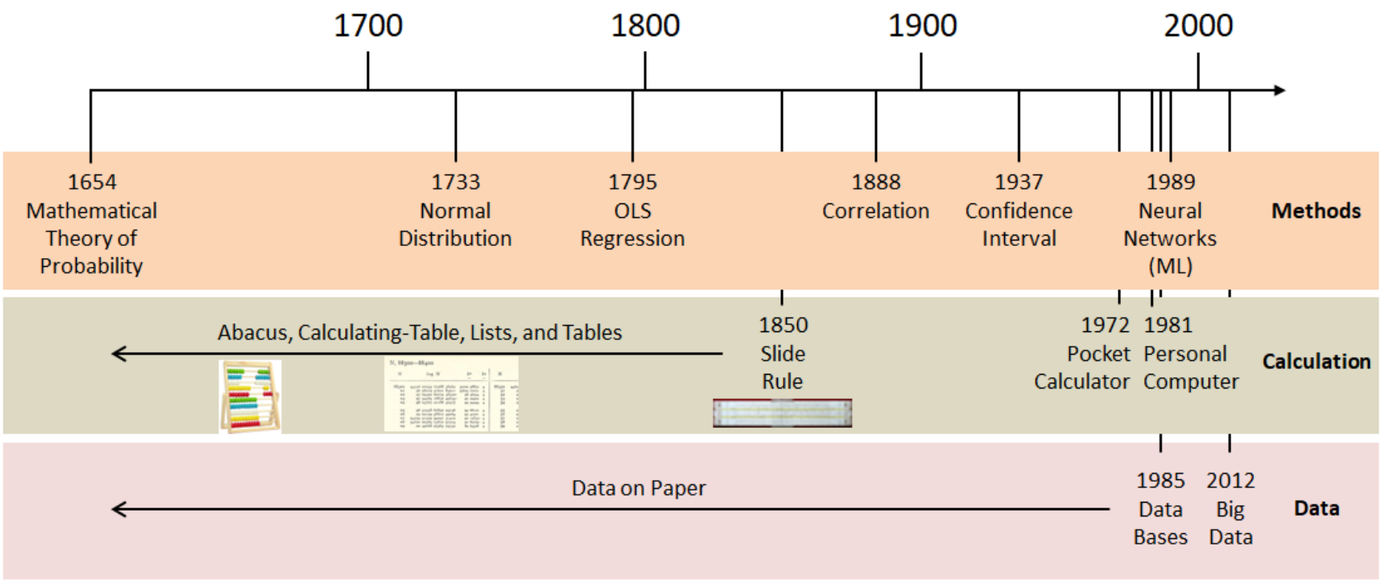
Traditional Analytics Process:



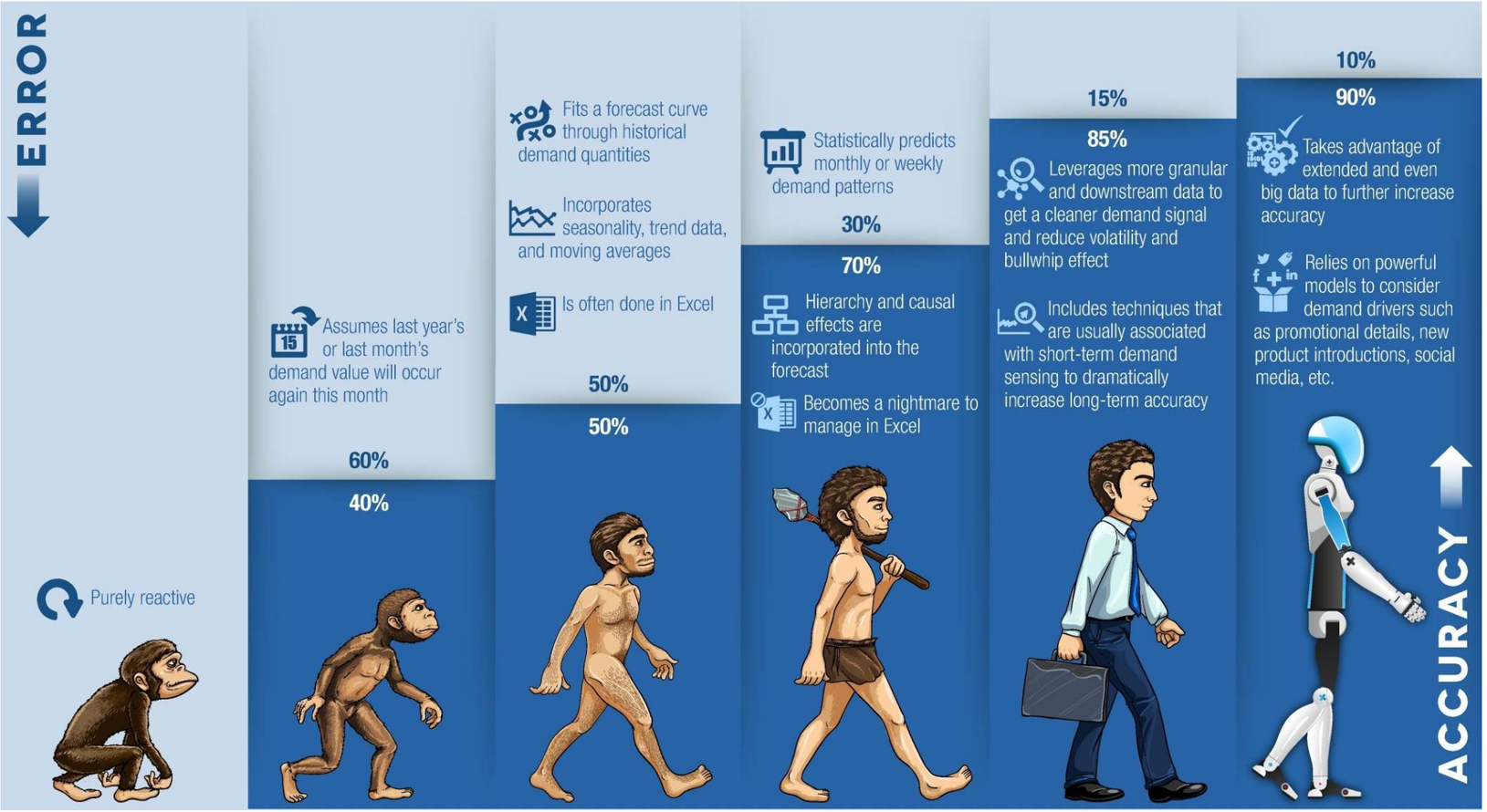
Modern Analytics Process:



From the Past to the Present

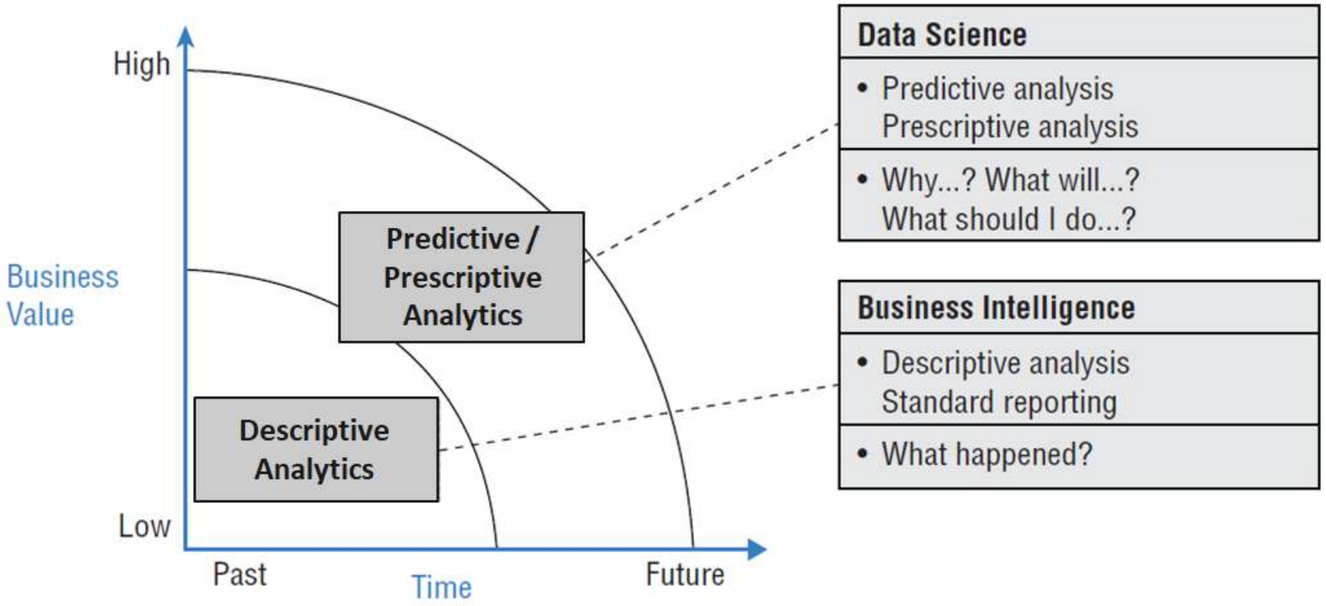


Evolution of Analytics



Source: http://juxt.pro/blog/posts/machine-learning-with-clojure.html

Types of Analytics (I)



Source: Schmarzo, p. 88

Types of Analytics (II)

Descriptive

- What happened in the past?
- Example: Report the profits of the last years.

Predictive

- Find patterns and relationships in data and use them for prediction.
- Example: Find the functional relationship between price and demand.

Prescriptive

- Use the predictions to make decisions.
- Example: Set the best price to optimize the profit.

TODO: the same analytical method may be applied for different purposes. maybe even switch to “Analytical purposes”?

Types of Analytics (III)

What happened? (descriptive / BI)	What will happen? (predictive analytics)	What should I do? (prescriptive analytics)
How many widgets did I sell last month?	How many widgets will I sell next month?	Order **5,000** units of Component Z to support widget sales for next month.
What were sales by zip code for Christmas last year?	What will be sales by zip code over this Christmas season?	Hire **Y** new sales reps by these zip codes to handle projected Christmas sales.
How many of Product X were returned last month?	How many of Product X will be returned next month?	Set aside **$125K** in financial reserve to cover Product X returns.
What were company revenues and profits for the past quarter?	What are projected company revenues and profits for next quarter?	Sell the following product mix to achieve quarterly revenue and margin goals.
How many employees did I hire last year?	How many employees will I need to hire next year?	Increase hiring pipeline by **35%** to achieve hiring goals.

Source: Schmarzo, p. 13

TODO: introduce CRISP-DM ??

Illustrate the multitude of models (here or in the “new algorithms” section?)

Discipline	Model Culture	Typical Models
Statistics	Probabilistic inference	Regression, GLM, Bayesian
Econometrics	Causal modeling	IV, panel models
Computer Science	Algorithmic learning	Trees, SVM, NN
Operations Research	Optimization	LP, MIP
Management Science	Decision modeling	Stochastic optimization
Engineering	Signal processing	State space, filtering
Complex Systems	Simulation	ABM, DES

-> highlight what we focus on. Mention that not all disciplines focus on the “business understanding” and using insights to support business decisions. Oftentimes, the goal may also be to understand and explain, not to decide (prescribe).

Statistics vs. Data Analytics

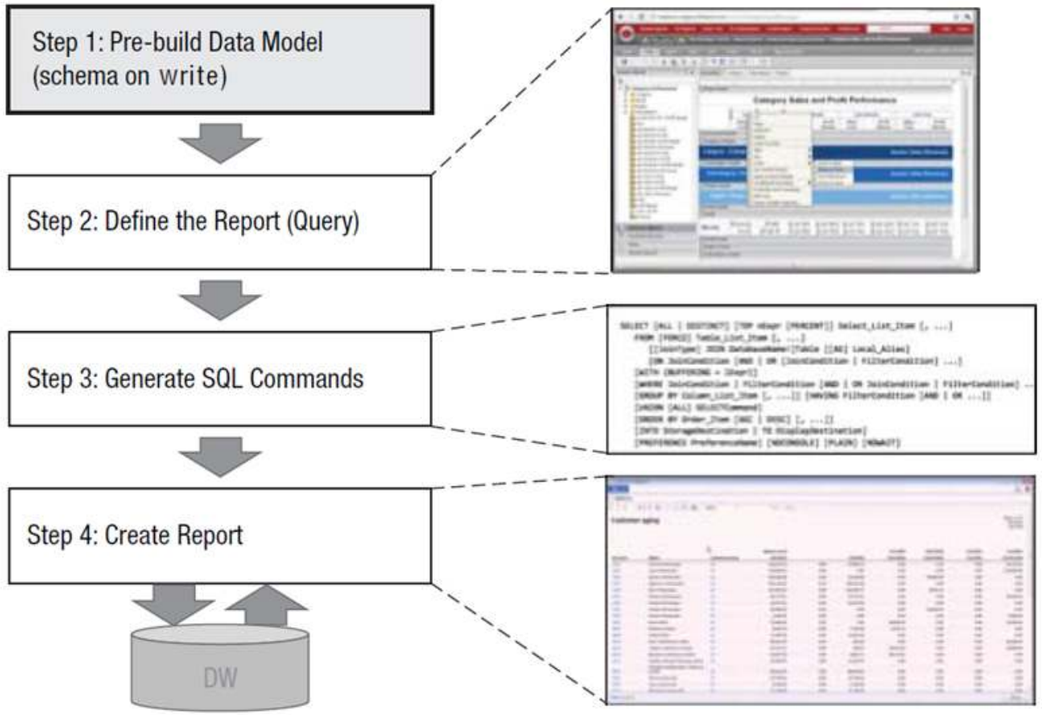
Statistics

Goals: Explanation, Forecasting

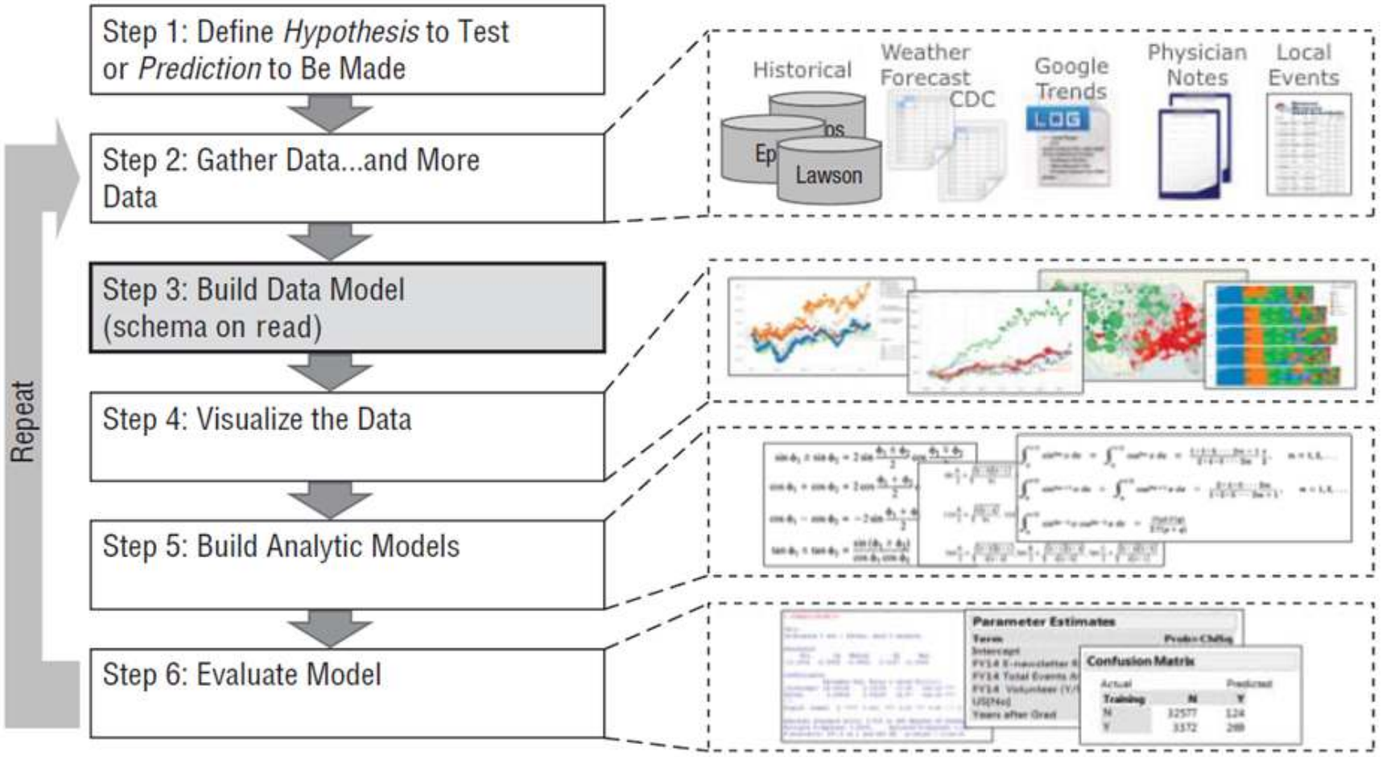
Methods: Statistical Methods

Process: Data ⇨ Method ⇨ Model ⇨ generalizable? (Statist. Tests)

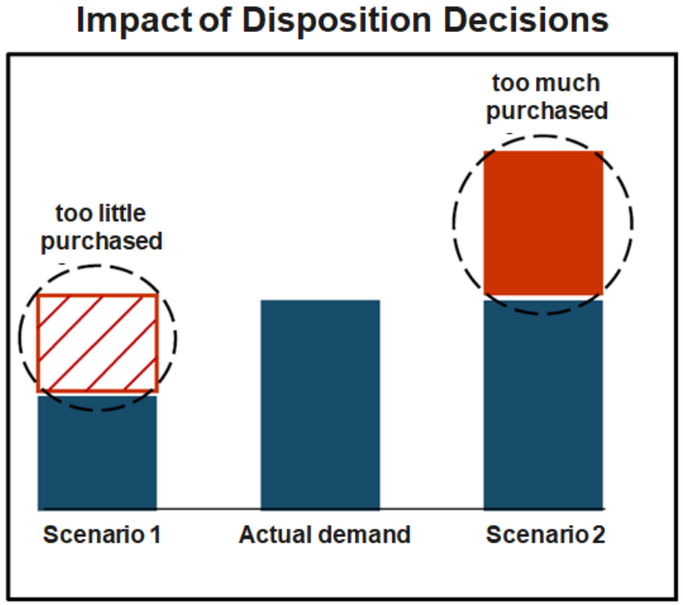
Descriptive Analytics Process



Predictive Analytics Process



Example Inventory Planning



Scenario 1: too little purchased

- Loss of turnover
- Increased procurement costs
- Customer annoyance
- Increased contact costs call center

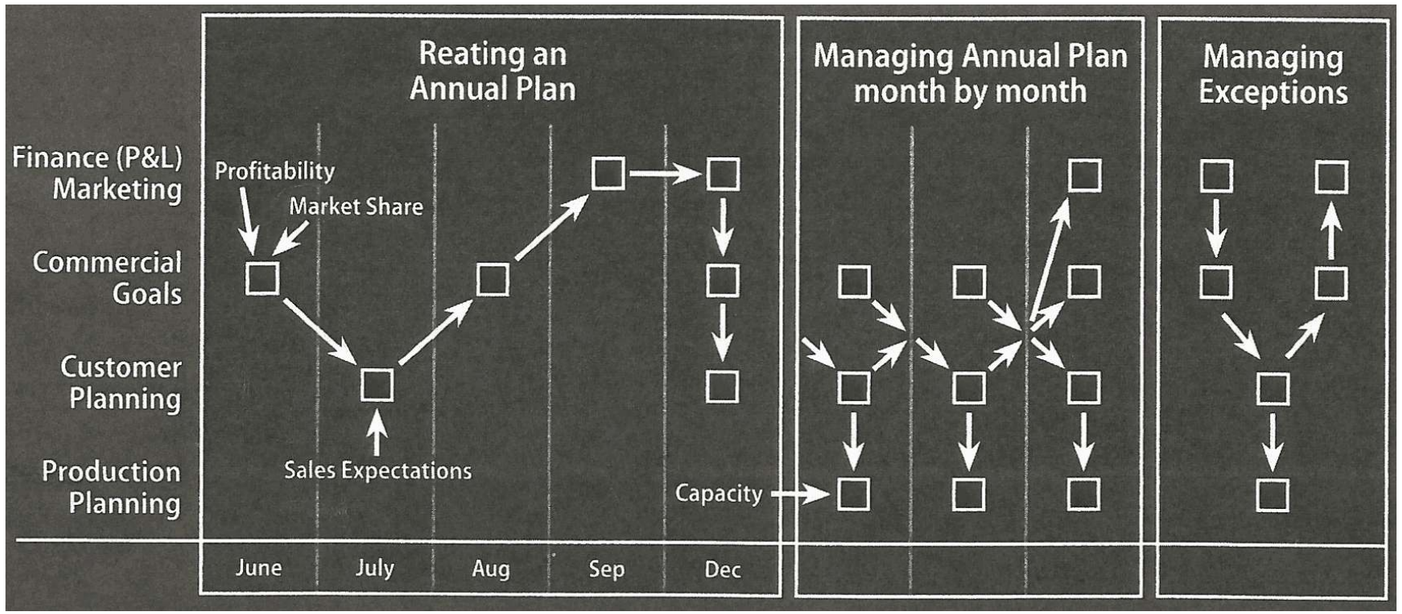
Scenario 2: too much purchased

- Increased storage costs
- Capital lockup
- Utilization losses

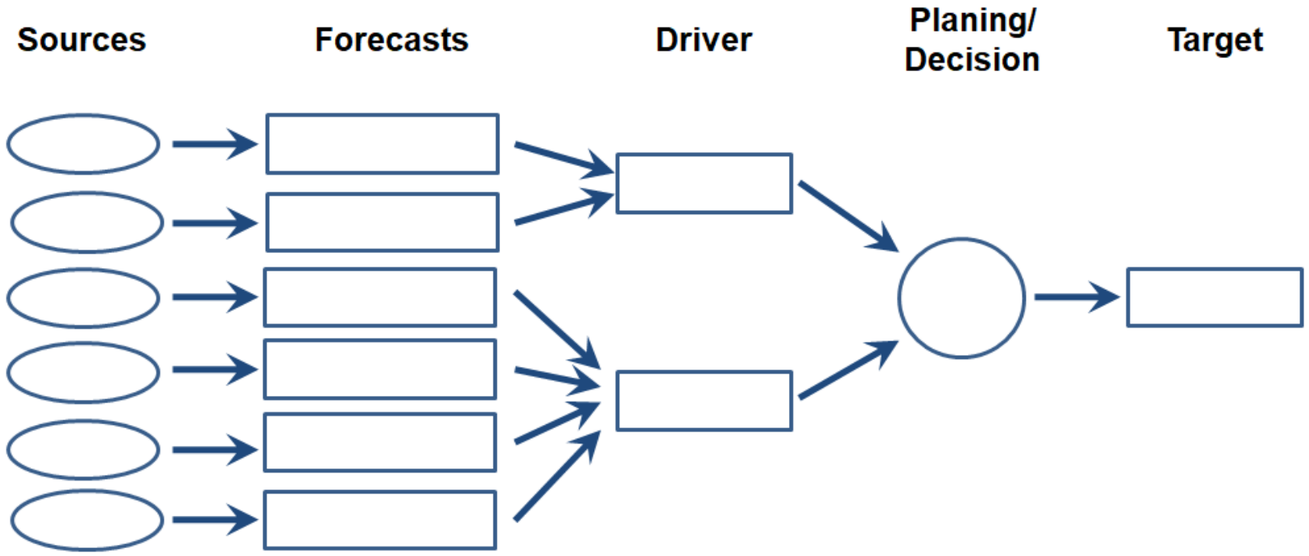
On the basis of more than 300 million data records per week, Otto makes more than one billion forecasts per year on the sales development of individual articles for the next days and weeks. Such forecasts allow Otto to reduce its own inventories by up to 30% on average.

Source: http://www.bvl.de/misc/filePush.php?mimeType=application/pdf&fullPath=/files/441/442/777/1015/DLK12_C3-3_Praesentation_Stueben.pdf

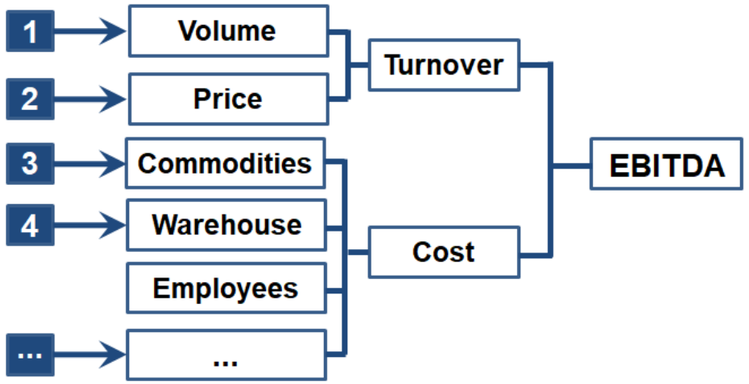
Classical Corporate Planning



Example Driver-based Planing (I)



Example Driver-based Planing (II)



1. Sales forecasts based on market and social media data
2. Automated pricing based on market and competitor analyses
3. Early detection of price changes and adjustment of purchasing behavior
4. Optimization of inventory management based on customers' current purchasing preferences

Summary

🎯 **Learning Objective**

- **Illustrate** how modern data analytics capabilities are enabled by data availability, advances in computing power, new algorithms, and modern analytics processes.
- **Distinguish** between descriptive, predictive, and prescriptive analytics.
- **Explore** the Python and Jupyter analytics ecosystem (*exercise*).